

Quiz 2 — Cleaning and Counting

M 375T Mathematical Foundations of Machine Learning — Fall 2026

15 minutes · closed book · no devices · same skills as Practice 2 & 3

Name: _____ UT EID: _____

1. A feature has sample mean $\hat{\mu} = 6$ and sample standard deviation $\hat{\sigma} = 2$. We standardize it:
 $z = (x - \hat{\mu})/\hat{\sigma}$.

(i) What value does the data point $x = 8$ map to?

(ii) What are the mean and the standard deviation of the standardized feature?

(iii) Which original value x maps to $z = -1.5$?

[4 pts]

Answer:

2. Read the code, then answer in words and numbers:

```
U = rng.uniform(size=(50_000, 100))
k = (U < 0.5).sum(axis=1)
print(k.shape)
print(np.mean(k == 50))
print(np.mean(k) / 100)
```

(a) What does the first print statement output, and — in one sentence — what does each entry of `k` count?

(b) The second print estimates a probability. Which one? Give the exact expression for it (no need to evaluate).

(c) Approximately what does the third print output, and why? [6 pts]

Answer:

Total: 10 points. Lowest two quiz scores are dropped at the end of the semester.